Algorithms in Bioinformatics

Spring 2026

Project 1, week 10

The goal of this project is to use the k^{th}-order Markov Model to classify a set of short genomic sequences. Each short sequence will be classified to one group, which is represented by a genome in a reference genome data set. A file containing short genomic sequences can be found at /lustre/share/class/BIO3501/proj1/reads.fa. The 10 genomes can be found under the directory /lustre/share/class/BIO3501/proj1/genomes/. A test set has been created for you: 20,000 short reads are given in /lustre/share/class/BIO3501/proj1/test/test.fa. The mapping of aligned genomes and the short sequences listed in test.fa is given in a file called seq_id.map file. You can use these files to test your own script.

You are expected to include in your report the implementation of your k^{th}-order Markov Model method together with the following results. Your scripts need to be handed in as supplementary materials.

1. the total number of short sequences in the file reads.fa; (10 points)
2. (*) the number of reads that can be assigned to a genome in the genome dataset (with whatever criteria you use); (10 points)
3. (*) the number of reads that can't be assigned to any genome in the genome dataset; (10 points)
4. the number of groups with at least j short sequences assigned, where j = 1, 5, 10 or 50. You can describe the results in a table similar to the one below. (20 points)

Minimum number of short sequences in a group	Number of groups
1	
5	
10	
50	

5. for those groups with at least 10 short sequences assigned, list the total numbers of short sequences assigned to those groups. Then you can draw a pie chart to show the relative frequency of each group. (10 points)
6. For k=3-20, which k value gives the best classification result? Please give your own criteria for "good" result, and explain why this k value gives the best result. (15 points)
7. Some researches have shown that combining different orders of Markov Models may improve the classification result. Please try different combinations of two or three different k^{th}-order Markov Models, and compare the sequence classification results to that of the best k^{th}-order model to see if there is a combination method performing better than all single-order Markov Model. (25 points)

This is a small-scale research project using the knowledge learned from your course. You need to give detailed information about each step in your analysis so that your result can be reproduced by others if it is needed.

Students are encouraged to form a team of two or three (at most three), and submit a report for the whole team instead of a report for each individual. However, you have to describe the contribution of each individual in the last paragraph of the report if you choose to work as a team. The lecturer may pick some of you to give a short presentation about your project.

(*Methods other than the k^{th}-order markov model are also allowed to finish this project. If you do decide to use a method other than the k^{th}-order markov method, you may omit question 7. However, you have to answer these questions started with * and compare your results with the results of another group using k^{th}-order markov model.)

Bonus

If you report how you use GPUs to do metagenomic sequence classification, and give the speedup compared to your CPU version, you will get a bonus of 10 points.

Turning in your project work

Submit your result files (all in one file and name it as Yourname_yourID_proj1.doc or pdf, please) via the course website. The project report should be handed in before 23:59 June 15, 2026, one day before the next class on June 16, 2026.

The result file(s) should include your project report and your source codes for the project. Please put your source codes separately so that we can run it directly. Also, please give the command lines to generate your results in the project report so that others can reproduce your results.

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独立作业承诺：（请选择一个， 并签名）

本人，__________，保证本次作业 hw1

☐ 由自己独立完成；

☐ 和__________同学讨论后，由自己独立完成。
讨论内容包括____。

☐ 使用了 AI_______（版本_______）后，由自己独立完成。使用 AI 的详细过程描述已经放在作业报告文件____________________________中。

☐ 和__________同学讨论，并使用 AI_______（版本_______）后，由自己独立完成。讨论内容包括____；AI 使用的详细过程描述已经放在作业报告文件____________________________中。

签名

时间　　　年　月　日